Application Building Steps

Building HTML Pages

Three HTML pages are created and stored inside the **templates/** folder to build the frontend of the application:

• **home.html:** Acts as the landing page, provides a project overview, and includes a navigation button to access the prediction form.

• **index.html:** Contains the input form where users enter applicant financial and demographic details required for prediction.

• **result.html:** Displays the final prediction result showing whether the credit card application is approved or rejected.

Building the Python Script (Flask App)

• The **app.py** file is used as the main backend script of the Flask application.

• It handles routing, model loading, input processing, prediction generation, and result rendering.

• The trained model is loaded using **Pickle** or **Joblib**.

• Flask routes are created to display the home page, prediction form, and result page.

• When the user submits the form, values are processed and passed to the model using the `predict()` method to generate the final approval prediction.

Data Pre-processing Steps

Here are the tasks for **Epic 3: Data Pre-processing**:

• **Drop Duplicate Features**

• **Handling Missing Values**

• **Data Cleaning and Merging**

- **Feature Engineering**
- **Handling Categorical Values**

Data Analysis Phases

Data Analysis Steps

- **Univariate Analysis**
- **Multivariate Analysis**
- **Descriptive Analysis**

Read Dataset Commands

Read the Dataset Commands

- **Load CSV File:** `app = pd.read_csv('application_record.csv')`
- **Load Credit Record:** `credit = pd.read_csv('credit_record.csv')`
- **View Top Rows:** `app.head()` (Returns top n rows, 5 by default)
- **Check Dimensions:** `shape`
- **Dataset Summary:** `info()` (Displays column names, data types, and missing values)

Credit Card Project Libraries

Required Python Libraries & Packages

- **NumPy**
- **Pandas**

• **Matplotlib**

• **Seaborn**

• **Scikit-Learn**

• **XGBoost**

• **Flask**

Credit Card Approval Prediction - Prerequisites

Credit Card Approval Prediction - Prerequisites

Here are the powerful Python-based tools, libraries, and resources required for data processing, machine learning, visualization, and deployment in this project:

• **Anaconda Navigator:** A desktop graphical user interface used to launch applications and manage Python packages and environments for data science. Download Anaconda(https://www.anaconda.com/download)

• **PyCharm:** An Integrated Development Environment (IDE) specifically designed for Python programming, offering code analysis and debugging tools. PyCharm Website(https://www.jetbrains.com/pycharm/)

• **NumPy:** Core library for numerical computing in Python, providing support for large, multi-dimensional arrays and mathematical functions. NumPy Documentation(https://numpy.org/doc/stable/)

• **Pandas:** Data manipulation and analysis library offering data structures and operations for manipulating numerical tables and time series. Pandas Documentation(https://pandas.pydata.org/docs/)

• **Scikit-learn:** Machine learning library featuring classification, regression, and clustering algorithms like support vector machines and random forests. Scikit-learn Documentation(https://scikit-learn.org/stable/)

• **Matplotlib:** Comprehensive library for creating static, animated, and interactive visualizations

in Python. Matplotlib Documentation(https://matplotlib.org/stable/)

• **Seaborn:** Data visualization library based on matplotlib that provides a high-level interface for drawing attractive statistical graphics. Seaborn Website(https://seaborn.pydata.org/)

• **Flask:** A lightweight WSGI web application framework used to build web APIs and deploy machine learning models. Flask Documentation(https://flask.palletsprojects.com/)